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IM-1 PRE-MISSION LANDING SITE EVALUATION

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One of the first and most critical challenges when designing a lunar surface mission is landing site selection. The candidate landing sites have bearing on the launch window, potential landing dates, vehicle trajectory (and DV), and landed attitude. Addressing the problem requires a team effort, from mission designers to planetary scientists to vehicle subsystem domain experts and software developers. This paper outlines the team workflow that was developed to choose Intuitive Machines IM-1 landing site for the Nova-C class Odysseys lander. From initial science driven choices provided by NASA, ASU, and IM, to engineering concerns, the IM-1 case study is examined along with the tools and data utilized in the landing site selection process.

INTRODUCTION

Intuitive Machines (IM) recently met its goal of softly landing near the south pole of the Moon in February 2024 with its IM-1 mission. This was the first time in 50 years the United States had landed on the Moon, as well as the first time a private company was able to accomplish this feat. The mission was catalyzed by a NASA Commercial Lunar Payload Services (CLPS) contract which aims to further explore the lunar surface in preparation for human missions to the Moon. While on the surface, the Nova-C vehicle was able to collect crucial environmental data. The success of this mission is not only a step towards laying the foundation for a sustainable presence on the lunar surface, but also represents a pivotal moment in the ongoing narrative of space exploration.

While a lunar mission's landing site is the final step in its timeline, the selection of a landing site must be one of the first decisions made when designing a lunar lander mission. This selection process must take in many considerations and requirements as each potential site influences the entire mission and has a large impact on mission success. The launch window comes from the landing date, which in turn is based on the relative elevation of the Sun and Earth at the landing site. Once the launch and landing dates are chosen, the trajectories for each launch/landing date pair are created. The delta-velocity (DV) for the mission comes out of the trajectory design process and the attitude timeline for the mission is generated. Surface operations are planned once the landed orientation of the vehicle is defined by the attitude timeline. This entire mission design process relies on the landing site selection, which means establishing a meticulous selection process is crucial to mission success.

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Figure 1. Picture taken by Nova-C lander from the lunar surface.

The list of requirements taken into account during the landing site selection process for the IM-1 mission came from three main sources: NASA, IM,and the Nova-C lander system. According to the primary requirements levied by NASA, the landing site had to be in the South Pole region. The Nova-C lander requirements were predominantly focused on power generation and communication system performance during surface operations. Normally, payloads are another driver of requirements, however for IM-1, the payloads were agnostic to landing location and only required landing safely. IM set the mission length requirement to provide an adequate amount of time for surface operations and set the parameters for landing safety. Further discussion on the requirements that constrained the landing site selection process is found in subsequent sections.

The landing site selection team utilizes open source tools developed by NASA's Navigation and Ancillary Information Facility (NAIF) and Applied Coherent Technology (ACT) in conjunction with tools created in-house. The ACT Quickmaps tool set are web-based mapping tools that aid in initial analyses of potential landing sites and confirming results found by IM tools. NAIF's "SPICE" (Spacecraft, Planet, Instrument, C-matrix, Events) system provides functions relating spacecraft to planetary bodies, which are utilized in IM tools. As a final analysis, the Landing Site Evaluator (LSE) tool was developed by IM to fill in gaps outside sources possessed as well as present analysis data in a comprehensive manner. The LSE tool employs the use of Digital Elevation Models (DEMs) developed by Arizona State University (ASU) using Lunar Reconnaissance Orbiter (LRO) data, along with planetary ephemerides from the SPICE toolkit as data sources when performing its analyses.

The landing site selection process begins with a requirement region where the mission must land.

The tools and data sources described above search the entire region for smaller areas of interest, and inside those areas of interest possible landing locations are found. From this list of possible locations, the site that best fits within the mission requirements is chosen as the intended landing site (ILS). Further analyses are conducted to find the rise and set times of the Sun and Earth, which bound the mission landing time. From there, the trajectory is generated. The final step in the process is ensuring the approach trajectory is over lit terrain and the landing uncertainty ellipse is lit and free of hazards.

This paper is arranged into five sections. The first will discuss the requirements that constrained the landing site selection process, including NASA, system-level, and mission requirements. The next two sections will review the tools and data sources exploited by the landing site selection team. In the fourth section there is a brief description of how the LSE tool was developed and what gaps it filled in the analysis. The final section is a case study of the landing site selection process applied to the IM-1 mission.

MISSION REQUIREMENTS OVERVIEW

Landing site selection requirements came from several sources, including NASA, the Nova-C lander, and the mission definition created by IM. The requirements levied by NASA were the primary boundaries used when first searching for an advantageous region to land in. NASA requested the mission land in the South pole region, far from any possible future CLPS mission sites.

From here, the decision of landing site was up to the IM team. The selection of a site in this region was defined by two main mission parameters: landing safety and mission length. The intended site must meet certain criteria to ensure vehicle safety and operation during and after landing. The site needed to have a low enough slope to avoid tipping, and be free of hazards, such as rocks or craters, that would also cause the vehicle to tip upon landing. As for the second requirement, the minimum mission length was set by IM to allow adequate time for all surface operations.

Nova-C was designed and constructed by Intuitive Machines as the answer to CLPS' contract for a lunar lander. The definition of mission length was derived from Nova-C's system requirements. The lander's power system was reliant on power generated by the solar arrays, as the batteries were not sized to power the vehicle for long periods of time. The thermal system also relied on the Sun for heat since the lander was not rated for the low temperatures of lunar night. Due to these constraints from the power and thermal subsystems, the Sun had to be above the terrain for the entire mission. Earth's elevation also had to be above the local terrain to allow for effective communication with Earth-based ground systems. Mission length was therefore defined as the length of time after landing both the Sun and Earth were above the horizon.

The vehicle relied heavily on the navigation system during powered descent as the vehicle operated autonomously after powered descent insertion (PDI) through landing. Therefore, the guidance and navigation (GNC) team required the approach trajectory be well-lit enough for the terrain camera to see the lunar surface in order to aid with the localization of the vehicle during powered descent and landing. The GNC team also set the boundaries of the intended landing site area. Their landing uncertainty ellipse was a 2 kilometer by 1 kilometer ellipse with the major axis along the approach vector. This area needs to be clear of large slopes and hazards before accepting an intended landing site.

DATA SOURCE OVERVIEW

The Lunar Reconnaissance Orbiter (LRO) was launched by NASA in 2009 with the primary objective to make fundamental scientific discoveries about the Moon.[1] Onboard LRO there is a system of three cameras called the Lunar Reconnaissance Orbiter Camera (LROC). LROC consists of two Narrow Angle Cameras (NACs) and one Wide Angle Camera (WAC). Images produced by the NACs provide high resolution shadowing data that are used by IM to analyze the illumination of potenital landing site areas.

LRO also has the Lunar Orbiter Laser Altimeter (LOLA) which takes measurements of surface height with a vertical precision of 10 cm and an accuracy of 1 m.[2] LRO's polar orbit has allowed LOLA to collect high resolution data in the region of interest for the IM-1 mission. Surface height data from LOLA has since been amalgamated into various Digital Elevation Models (DEMs).

DEMs created from the LRO data were the main source of elevation for the landing site selection team. Up to the time of the mission, LOLA provided the highest resolution and most accurate polar terrain models.[2] The 5 meter per pixel DEMs allow analysts to investigate areas of interests for large hazards like boulders and craters. Potential landing sites can be found that are in areas free of such hazards, which greatly increases the possibility of success for the mission. On the contrary, any hazards that are smaller than 5 meters diameter will be missed by the DEM. This is especially concerning considering the required hazard minimum of the landing site is 2 meters diameter. Any potential hazard that isn't seen in the DEM that is larger than this requirement might lead to tip over at landing.

IM's LSE tool uses the topography data the South pole DEMs provide to generate relative elevation maps around a landing site. The terrain relative elevation data is then used to refine landing dates in a potential landing window. The process of how the LSE tools uses DEM data is discussed further in the IM Landing Site Evaluation Tool Section.

As discussed earlier in this section, LRO's polar orbit yields high resolution terrain maps of the South Pole region. It also allows NAC to capture high resolution lighting images of the region. NAC data is used extensively by the IM team. NAC images with lighting as similar as possible to the expected lighting at landing is used as ground truth to confirm the lighting conditions predicted by the LSE tool.

Another data source utilized in the selection of IM-1's landing site is NASA's Navigation and Ancillary Information Facility's (NAIF) Spacecraft, Planet, Instrument, C-matrix, Events (SPICE) information system. SPICE was developed as a pre-mission planning system for NASA scientists, with scientific observations from space-borne instruments and planetary exploration missions.[3] The SPICE system uses ancillary data to help analysts compute the relative geometry between planetary bodies and spacecraft. This ancillary data is helpful in all mission stages, from mission concept development through the post-mission data analysis phase.

The landing site selection process applies SPICE's ancillary data to the task of calculating the Sun and Earth's elevation at a potential landing site. As discussed in the previous section, the Sun and Earth's relative elevation above the terrain defines the length of the surface mission. The accuracy of the elevation calculation, therefore, can significantly alter the landing site selection process. The calculations provided by SPICE are trusted to output the most accurate data possible.

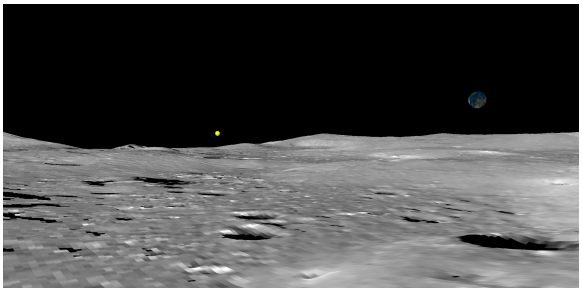

Figure 2. Example of using Quickmaps 3D to find Sun and Earth visibility windows at arbitrary time and location.

LANDING SITE EVALUATION USING OUTSIDE TOOLS

ACT's QuickMap is a powerful GIS-like web based tool that permits quantitative analysis of multiple lunar data sets. These data sets include the LROC, NAC, and WAC products as well as topography and slope maps. One valuable function is the ability to ingest user-developed Geotiffs directly into QuickMaps, which will be discussed later in this section.

The IM team utilizes the elevation and slope maps provided by QuickMaps early in the landing site selection process. The maps easily allow the team to visualize high slope areas and larger hazards like craters when investigating a large region. Further refining the region into potential landing sites is supported by the tool's practice of providing the highest resolution data of the area being analyzed. Once the intended landing site is selected, the team next utilizes QuickMaps' shadow tools to examine the site's lighting.

QuickMaps includes multiple shadow tools that display the illumination of the lunar surface at different epochs. One of these tools, called Quickmaps 3D, allows the user to see a landed view of the lunar surface at a desired location. Once at this view, the Sun and Earth are shown at their relative orientations over the lunar terrain. The epoch can be changed to show the what their orientations would be at any time. The IM team uses the landed views the tool generates to identify the main Earth and Sun visibility windows each month. Landing windows are further refined using the LSE tool and QuickMaps' other shadow tool, QuickMaps 2D.

Quickmaps 2D takes in a time, definition of illumination (as in what part of the Sun defines rise/set), start UTC time, coordinates to center the analysis, DEM resolution used, height offset from surface, and the maximum distance to check for objects blocking the Sun away from the center point. The tool generates a GeoTIFF file, which is uploaded into the Quickmaps main tool. Analysts can then see the illumination around the point of interest. The landing site selection team uses this data as a qualitative test to further refine the landing windows found by using QuickMaps 3D. The dates derived from this analysis are then compared with the output from the LSE tool and NAC data, which is considered as truth. This comparison produces a set of landing days that are vetted by all three data sets to guarantee favorable lighting conditions.

Quickmaps is a powerful tool that is critical to the landing site selection process. The data provided gives the team a better idea of areas that will meet all mission requirements in an easy to view visual. The visualization also provides clear comparisons of multiple potential landing sites or regions at once. The tool is also a great source to confirm conclusions made from the landing site evaluation tool and helps create a complete analysis of each potential landing site.

IM LANDING SITE EVALUATION TOOL

The initial driver for the landing site evaluation (LSE) tool stemmed from the lack of existence of a ready tool to assess several types of analyses. Though QuickMap allowed areas to be drawn and layers such as elevation and slopes to be calculated, there was no ready means of creating statistical data or ensuring that only an area of 1 km around the ILS was examined. Furthermore, in 2020, QuickMap did not possess the suite of capabilities it has today, i.e. QuickMap 2D (TerrainShadows) and QuickMap 3D (QTS-3D) did not exist. At that time, there was no means to calculate the mission surface duration requiring both sun and Earth to be above the terrain horizon with respect to an ILS.

IM at the time looked into two options concurrently. The first was to potentially hire a company to develop such a tool for us. A company called Lunar Station Corporation which has a software called MoonHacker promised to be able to deliver. The second was to develop our own set of tools within our Python ecosystem that the IM synthetic imagery team has started initiating. After a trial period of approximately five months, IM Flight Dynamics team made the decision to not proceed with a third-party vendor but instead to focus efforts on developing our own internal tools.

The LSE tool was built from a foundation which took initial inspiration and code from what the synthetic imagery team had developed at that time. The initial code was written in Python 3 and extensively used the gdal library to handle ingestion and processing of DEM data as well as SpiceyPy library to know where the sun position is. gdal is a powerful suite of terminal applications as well as scripting functions (in Python) frequently used in the Geographical Information Systems (GIS) field which can be used to manipulate, transform, read and write to digital elevation models as well as other geographical raster datasets such as slope maps. SpiceyPy is an open-source Python wrapper for the SPICE Toolkit, which is a collection of tools from JPL NAIF as noted earlier.

Since the purpose of the LSE was different from that of generating synthetic imagery, i.e. it was not meant to produce a photo-realistic landscape but rather to assess landing site viability, the tool was placed in a separate repository to continue development independent from existing synthetic imagery code base. From a high level, the LSE tool was built with three basic functionalities in mind. Additional capabilities have been added over the years for ease of usage, improved capabilities and graphics, and to support other analysis such as optical navigation but these three have remained as fundamental for choosing the landing site:

1. Generate heatmaps of the elevations and slopes around the landing site and provide such data in tabular form.
2. Plot out sun and earth positions relative to the intended landing site with respect to time after landing.
3. Identify when the spacecraft on surface will have power (from sun) and communications (from earth) post-landing.

In order to accommodate the three main functionalities, LSE has several different scripts and capabilities within scripts.

Landing Site Evaluator: DEM / Slope Map Reader

The first functionality in the LSE is to read in a DEM or slope map file and display a heatmap within a certain distance from a list of landing site coordinates in a landing site configuration file. This will also have an associated data in a csv format which has the raw data. The code does this

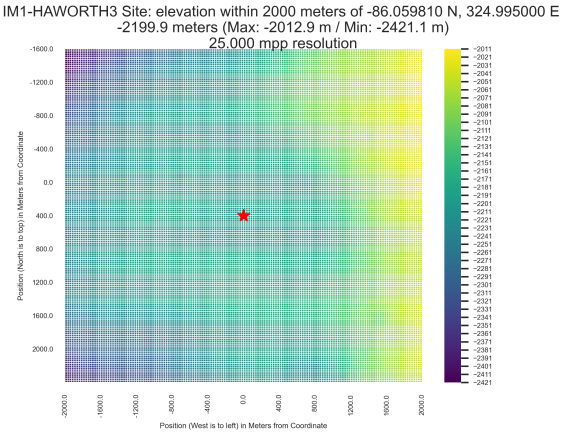

Figure 3. Example of an elevation map of Haworth 3 site

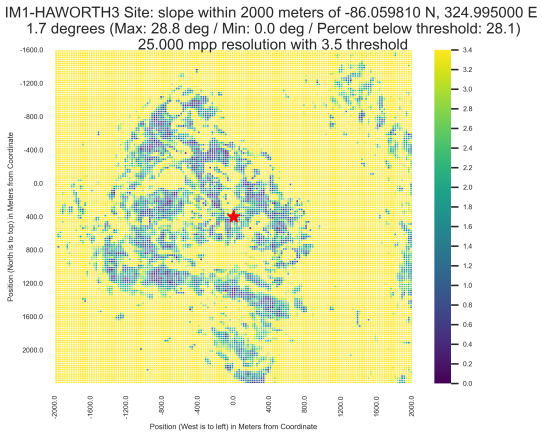

Figure 4. Example of a slope map of Haworth 3 site

by first reading in the DEM or slope map with gdal which is also able to read in the metadata for understanding the projection of the data. The output is a large 2D array of data points where each data point corresponds to either an elevation or a slope. The code is smart enough to understand what is elevation and what is slope depending on the numerical value of the data. If the data is a DEM, the code also checks and does any unit conversion necessary so the 2D array outputted is the elevation in meters above the mean lunar radius of 1737400 meters. The code then reads in a desired latitude and longitude from the landing site configuration file and transforms it into a corresponding pixel and line coordinate. Depending on the radius the user indicated, which for IM1 was 500 meters, the code divides this by the horizontal resolution of the dataset and "cuts out" that portion of the 2D array. For instance, a DEM with resolution of 10 meters per pixel will mean that 50 data points will correspond to 500 meters. Since the landing site is at the center, this means cutting out a 100 x 100 2D array. The corresponding heat map of appropriate size is then created. The "cut out" array's horizontal and vertical indices are transformed back into the corresponding latitude and longitude based on the metadata projection. The list of latitude and longitude along with each of its corresponding data (either elevation or slope) is then saved as a csv file with three columns of latitude, longitude and data. This is repeated for every landing site coordinate in the landing site configuration file.

For slope map, the latest version of LSE also allows the user to adjust threshold slopes. This then produces statistical results for what area is considered below the threshold of acceptable slope risk. As can be seen in Figure 4, 28.1% of terrain within 2 km of the ILS has slope greater than 3.5 degrees.

Sun Earth Azimuth Finder

The second functionality on determining the Sun and Earth positions at a landing site and how those evolve over time is critical for determining the expected power and communications profile during surface operations. This is the only functionality which does not use gdal but instead just uses SPICE and it is created by a script called the Sun Earth Azimuth Finder. Specifically, it loads in the Sun, Earth and Moon ephemeris over a given span of time with a particular time step. Given a ILS coordinate with latitude, longitude and altitude (optional and not considered important given the immense distance to Earth and Sun), the code calculates the azimuth and elevation angles of the Sun and Earth with respect to time. Azimuth angle is defined as ranging from 0 to 360 degrees with North being 0 degrees, East being 90 degrees, South being 180 degrees and West being 270 degrees. Elevation angle is defined as ranging from 0 to 90 degrees with 0 degrees flat along the horizon and 90 degrees being straight up above the surface. In the South Pole region, the Earth elevation angle varies from around 2 to 14 degrees while remaining mostly within 5 degrees of compass azimuth direction. The Sun elevation angle at the poles is very low and never rises above 6 degrees whereas in 1 lunar day (30 Earth days), it goes around 360 degrees in azimuth.

Landing Site Evaluator: Illumination Analyzer

On identifying how long a spacecraft sitting at an ILS will have direct view of the Sun and Earth, several strategies were considered on understanding whether the landing site will be illuminated (within view of the Sun) and observable (within view of the Earth). In 2020, the synthetic imagery team was still investigating the best software to create photo-realistic renderings of the lunar surface based on the optical navigation cameras and there was no ready solution available from that perspective. Furthermore, there was no research into how "illumination" from Earth will be modeled. For this reason, the LSE tool's algorithm to derive Earth and Sun visbility took a mathematical vector

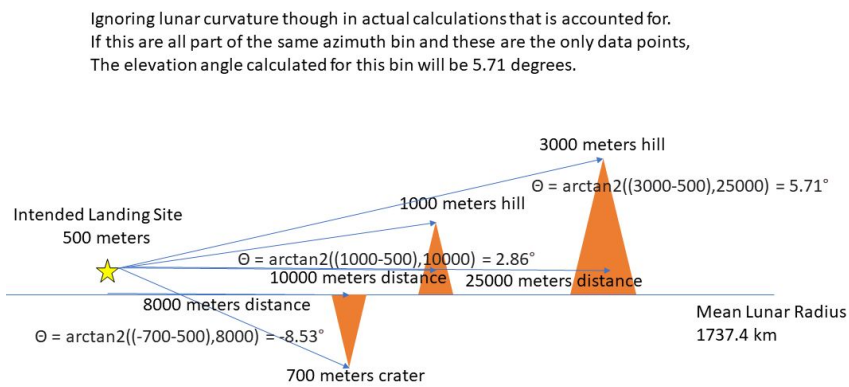

Figure 5. Drawing of how elevation angle for terrain is calculated.

approach instead of trying to build off what synthetic imagery was working on.

This tool is several tools operated in sequence. The primary unique analysis is to consider blockage by building a terrain angle map around the ILS with each local azimuth having a corresponding maximum elevation angle for terrain blockage. This is generated by an option called Illumination Analyzer within the larger LSE tool. The user defines which DEM to ingest and the furthest terrain it should consider for terrain blockage analysis.

The mathematics uses basic trigonometry by calculating the elevation angle with an arctangent where the y is the relative terrain height (the hill in the distance minus the height of the ILS) and the x is the distance to the terrain. For faster computation and considering that most calculations have x much greater than y, the small angle approximation can be optionally used where the angle is simply the delta height difference divided by the distance between the ILS and the feature. In addition, a small drop in angle due to lunar curvature assuming a perfect sphere is also accounted for. At the finest resolution level, this is done for each data point up to a distance away provided by the user. As long as it is the same azimuth away, any terrain that creates a greater elevation angle is considered the final blockage angle for that azimuth. However, the data could not be taken at face value due to the discrete nature of the data and also noise in the data.

The discrete nature of data stems from the fact that all DEMs have a fixed resolution and since south pole DEMs do not have equatorial projections. This leads to a situation where no terrain feature will have an azimuth direction that is exactly the same. This means that in practice, we approximate the same azimuth by placing them in bins that are considered close enough. For instance, consider a DEM that is 50 meters per pixel. A feature that is 150 meters away in the North direction (0 degrees azimuth) and another that is 200 meters away at 2 degrees azimuth will be considered together since 0 to 3 degrees azimuth may be considered close enough. An example is shown in Figure 5 depicting this scenario.

There was a series of trial and errors in tuning the calculator on the distance steps to do each trigonometric calculations and the bin size for each azimuth range to use to combine the data and then produce interpolated data. If the bin size is too small, there will be terrain blockages calculated that do not match with reality due to noise in the data. If the bin size is too large, the terrain blockage will be ”smeared out” and the average will not provide a realistic depiction of actual blockage. This terrain blockage data was extensively compared with QuickMap to validate that the

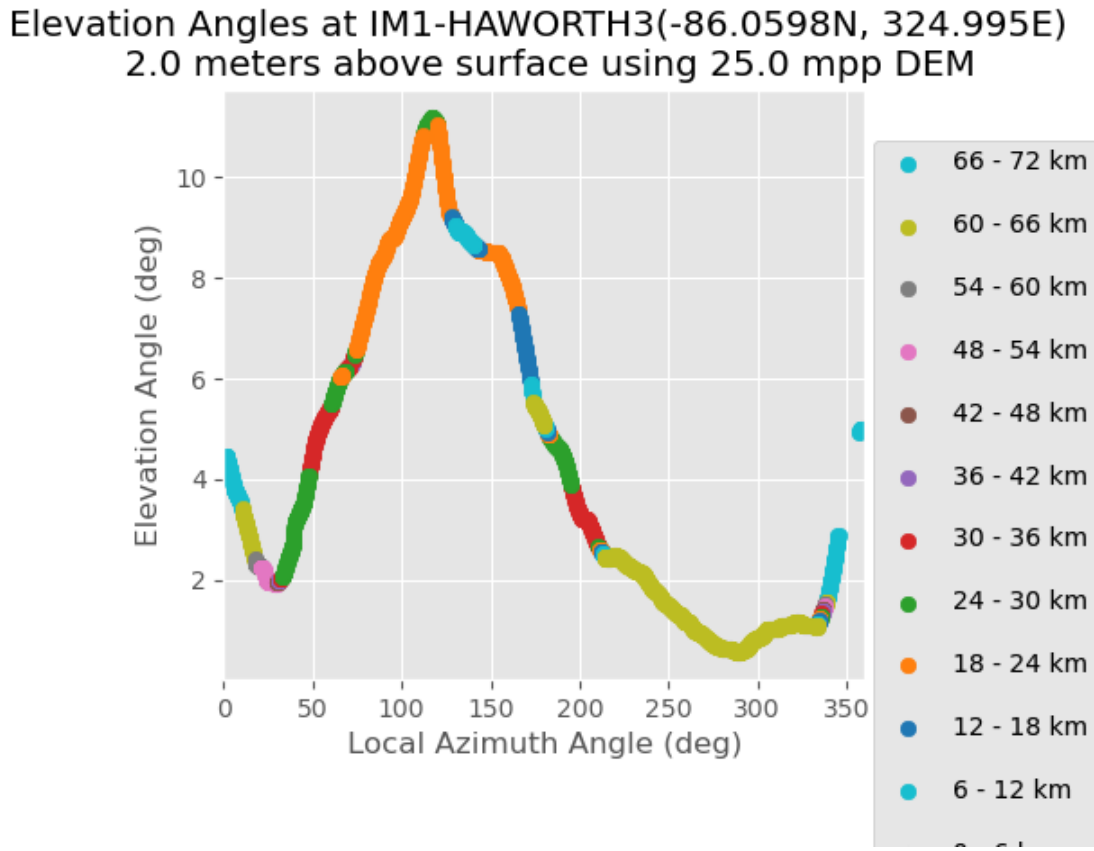

Figure 6. Example of the elevation angle for a site called Haworth 3

terrain blockages noted is accurate. Ultimately, the azimuth angle bin size chosen is 3 degrees. As a final means of ensuring that no outlier data is accepted, only elevation angles calculated that is between -10 and +15 degrees are accepted.

As a final step to ensure that sufficient points are provided all across the 360 degrees local azimuth, the data is interpolated in 0.12 degree azimuth points. The final output of the Illumination Analyzer is a csv file of local azimuth, terrain blockage elevation angles, and the obstacle distance creating this blockage. A terrain blockage plot is also generated as can be seen in Figure 6.

After the terrain blockage elevation angle data is created for an ILS, there's a separate post-processing script called Illumination Plotter to co-plot both the terrain blockage elevation angles data produced by the LSE Illumination Analyzer and the plot generated by the Sun Earth Azimuth Finder. The code also calculates and outputs the Sunrise and Earthrise time as well as the Sunset and Earthset time. Typically, a human analyst also double checks the data to ensure it's sensible.

IM-1 CASE STUDY

The IM-1 mission is used as a case study proving the efficacy of the landing site evaluation tool utilized in selecting an acceptable landing site. The use of outside tools, as well as the success of the mission will be used as markers for the accuracy and usefulness of the tool. The full process of selecting a landing site for the IM-1 mission is described below.

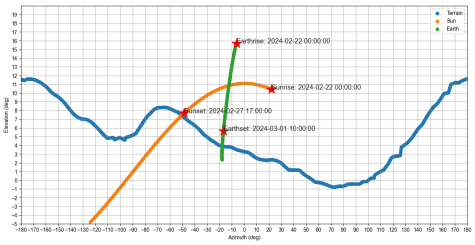

Figure 7. Example of output from LSE at arbitrary time and location.

As discussed in the Mission Requirements Overview section, the CLPS contract was focused on landing in the South Pole region of the Moon. Areas of interest inside this region were found and analyzed by the IM team. The areas were inspected for large hazards and slopes and illumination during desired landing windows. Most of these large areas were inspected using Quickmaps as a qualitative analysis of the area. Other teams such as GNC also gave inputs on the risk of landing in each area.

The area near Malapert crater came out to be the most viable landing area due to its flat topography, illumination during proposed mission windows, and low probability of hazards. Quickmaps was once again utilized to further refine the landing site search to a list of potential landing sites. The area around each site needed to be relatively low slope, with no large jumps in elevation, and no large hazards nearby.

The LSE tool analyzed the potential landing sites, examining their elevation, slope, and amount of illumination. High resolution LOLA DEMs provided the elevations and slopes, which were used in subsequent analyses. The main job of the LSE was to calculate the rise and set times of the Earth and Sun. For a deeper explanation of how the LSE works, refer to the IM Landing Site Evaluation Tool section. As discussed in the Mission Requirements Overview section, the vehicle could not land until both the Sun and Earth are above the horizon, and the mission ended once one body goes below the horizon. NAC imagery was compared with the LSE's rise and set times to confirm the accuracy of the times before moving forward with the analysis. The time at which the mission cannot land before, called No Earlier Time (NET), and the time at which the mission cannot land later than, called No Later Time (NLT), were backed out of the rise and set times. These times constrained the mission length and defined the landing time at each site.

The Intended Landing Site for the mission is supposedly the site that best meets the mission requirements, but the decision can be subjective at this point depending on customer and program wants. There are cases to be made about all sites, with some leading to shorter mission times but earlier landing dates, and others having higher slopes but longer mission durations. Whatever the decision-making process, the output was the program's current ILS.

The ILS along with NET and NLT was sent to the trajectory team. The team tried to create viable trajectories from Earth to this site, making sure constraints on the trajectories such as DV are met. Each trajectory's approach was checked by the landing site selection team to ensure the vehicle was traveling over lit terrain for the last portion of flight. If not, the trajectory was not considered viable. Depending on the amount of feasible trajectories, the ILS might be changed to another potential landing site. This process was be repeated until a site which produces enough viable trajectories was found.

CONCLUSION

The success of the IM-1 mission validated the landing site process created by IM. The tools and methods utilized by the process evolved throughout the IM-1 mission development. The result of the process refinement mitigated landing risks, lead to faster decision turn around and more accurate assessment of potential sites. The IM team is currently utilizing this process in the planning of future lunar lander missions, including IM's next lander IM-2. As the requirements for new sites become more stringent for future missions, and the number of candidate sites grows, the landing site selection process will use the fusion of tools and analyses currently developed to find the best candidate sites to ensure mission success.

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